

Training large language models (LLMs) as interactive agents presents unique challenges including long-horizon decision making and interacting with stochastic environment feedback. While reinforcement learning (RL) has enabled progress in static tasks, multi-turn agent RL training remains underexplored. We propose **StarPO** (State-Thinking-Actions-Reward Policy Optimization), a general framework for trajectory-level agent RL, and introduce **RAGEN**, a modular system for training and evaluating LLM agents. Our study on four stylized environments reveals three core findings. First, our agent RL training shows a recurring mode of **Echo Trap** where reward variability cliffs and gradient spikes; we address this with **StarPO-S**, a stabilized variant with trajectory filtering, critic incorporation, and gradient stabilization. Second, we find the shaping of RL rollouts would benefit from **diverse initial states, medium interaction granularity and more frequent sampling**. Third, we show that without **fine-grained, reasoning-aware reward signals**, agent reasoning hardly emerge through multi-turn RL and they may show shallow strategies or hallucinated thoughts.

Keywords: LLM Agents, Multi-turn RL

Website: <https://ragen-ai.github.io/>

Code/Environments: <https://github.com/RAGEN-AI/RAGEN>.